Fiscal Year:	FY 2015	**Task Last Updated:**	FY 06/19/2015
PI Name:	Hyers, Robert Ph.D.		
Project Title:	Unified Support for THERMOLAB - ISS, ICOPROSOL, and PARSEC		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	2
Comments:	Formerly with University of Massachusetts at Amherst.		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	09/01/2010	**End Date:**	11/15/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 11/15/2015 per NSSC information (Ed., 1/13/16) NOTE: End date changed to 9/30/2015 per NSSC information (Ed., 9/20/13) NOTE: Extended to 9/30/13 per PI, who also said it would be further extended (Ed., 8/5/13)		
Key Personnel Changes/Previous PI:	Robert Hyers is U.S. Team Member on ESA-sponsored projects, "Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC)," "Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS)," and "Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL)." Dr. Jonghyun Lee joined UMass Amherst full-time in June 2013. He had previously been a Tufts employee working on our collaboration on LODESTARS and PARSEC.		
COI Name (Institution):			
Grant/Contract No.:	NNX10AR71G		
Performance Goal No.:			

Performance Goal Text:

Task Description:

Funding is for Dr. Hyers's role as Team Member on the subprojects described below, associated with ESA-sponsored main projects as noted.
Main project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany

Subprojects:

1. Transformation Kinetics-ISS. Task description: Using results from thermophysical property evaluation and combining these with the Magnetohydrodynamic (MHD) convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the MSL-EML to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection.

2. Thermophysical Properties. Task description: The density, viscosity, and surface tension of the flight samples will be measured in flight for a range of temperatures. Density will be determined by the video metrology, and surface tension and viscosity by the oscillating drop method. This WP includes determination of parameter modifications required for these measurements as well as collection, reduction, and analysis of the resulting data. The resulting data will support the whole range of theoretical and modeling efforts in this proposal.

3. Modeling of Macroconvection. Task description: Magnetohydrodynamic (MHD) model of macroconvection in the samples for different combinations of experimental parameters. Evaluation of fluid velocity, shear rates, turbulence intensity, and their distribution for control of experiments and input to other theorerical and modeling efforts. Collaboration on development of parameters for other work packages to enable the use of convective state as an experimental parameter.

Main project: Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL); Science Coordinator is Dr. Dirk Holland-Moritz, Deutsches Zentrum für Luft- und Raumfahrt (DLR), Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany.

Subproject:

Modeling of macroconvection: Magnetohydrodynamic modelling of macroconvection in Ti-Zr-Ni samples for different combinations of experimental parameters. Evaluation of fluid velocity, shear rates, turbulence intensity, and their distribution for control of experiments and input to other theorerical and modeling efforts.

Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany

Subprojects:

Long Duration Microgravity Experiments - ISS. Evaluation of microgravity experiments for fluid flow investigations.

MODELLING A: Thermodynamics, Phase Selection. Modelling of nucleation under various flow conditions for steel alloys.

MODELLING B: Non-contact thermophysical. Modeling of magnetohydrodynamic convection and shear in 1-g, microgravity for 4-6 alloys. Modeling of Marangoni convection in 1-g ESL drops for comparison with microgravity experiments. Modeling of temperature distribution in 1-g and microgravity samples, 4-6 alloys. Input into design of experimental parameters for 4-6 Alloys.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

This project consists of modeling in support of three different flight experiments:
- PARSEC - Peritectic Alloy Rapid Solidification with EM Convection

- ThermoLAB - Thermophysical Properties of Liquid Metallic Alloys

- ICOPROSOL - Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order

In PARSEC, we are investigating the effect of fluid flow on the solidification path of certain alloys. Control of the solidification path would enable tailoring of the microstructure and properties of metal parts for specific applications.

In ThermoLAB, we are investigating the thermophysical properties of high-temperature materials. Many of these materials are used commercially. A better understanding of the physical properties will allow more efficient and more reliable production of metallic parts using these alloys. Other alloys are chosen because of scientific interest in their properties, which may illuminate fundamental physics related to the formation of bulk metallic glasses and quasicrystals.

In ICOPROSOL, we are investigating the nucleation and growth of quasicrystals, and the effect of atomic-scale order on the macroscopic properties of these alloys. This fundamental investigation may improve our ability to tailor the microstructure of metals for commercial applications.

Task Progress:	Highlights of Work Accomplished During Grant Period 2014-15. • Theoretically calculated the amount of mass evaporation of the FeCrNi sample tested during FCE1.2 to support the subsequent Batch1 processing of ThermoLAB and PARSEC samples. • Supported the magnetohydrodynamic simulations for the prediction of phase selection during solidification for PARSEC. • Supported the measurements of thermophysical properties of Fe-Co systems including the mass evaporation calculation for PARSEC. • Ms. Jie Zhao is investigating the influence of oxygen on the thermophysical properties of molten Zirconium for her Ph.D. thesis project in collaboration with NASA MSFC. • Attended the Herlach symposium and MSL-EML Investigators Working Group meeting, Cologne, Germany, September 2014. • Attended MSL-EML Investigators Working Group meeting, Cologne, Germany, May 2015.
Bibliography Type:	Description: (Last Updated: 03/30/2026)
Articles in Peer-reviewed Journals	Lee J, Matson DM. "Prediction of mass evaporation of Fe50Co50 during measurements of thermophysical properties using an electrostatic levitator." Int J Thermophys. 2014 Oct;35:1697-704. https://doi.org/10.1007/s10765-014-1662-9 , Oct-2014
Articles in Peer-reviewed Journals	Lee J, Xiao X, Matson DM, Hyers RW. "Numerical prediction of the accessible convection range for the electromagnetically levitated Fe50Co50 droplet in space." Metallurgical and Materials Transactions B. 2015 Feb;46(1):199-207. http://dx.doi.org/10.1007/s11663-014-0178-9 , Feb-2015
Articles in Peer-reviewed Journals	Lee J, Hyers RW. "Numerical estimation of convection in a molten zirconium droplet processed by electromagnetic levitation in microgravity." International Journal of Microgravity Science and Application. 2015 Jun;32(1):320105. https://doi.org/10.15011/jasma.32.1.320105 , Jun-2015
Articles in Peer-reviewed Journals	Lee J, Rodriguez JE, Hyers RW, Matson DM. "Measurement of density of molten Fe-Co alloys using electrostatic levitation." Metall Mater Trans B. 2015 Dec;46:2470-5. https://doi.org/10.1007/s11663-015-0434-7 , Dec-2015